

STAGE 3

Setting paths:

```
cd /home/user01/springboot-25-log4j-structural/spring-boot-v2.5.0
```

```
R="$PWD/reports_codeql_springboot"
```

```
Q="$R/queries"
```

```
DB="$R/springboot-codeql-db"
```

```
mkdir -p "$R" "$Q"
```

```
rm -rf "$DB"
```

Create a database:

```
codeql database create "$DB" \
```

```
--language=java \
```

```
--source-root "$PWD" \
```

```
--build-mode=none
```

Query pack:

```
cat > "$Q/qlpack.yml" <<'EOF'
```

```
name: local/springboot-log4shell-reachability
```

```
version: 0.0.1
```

```
dependencies:
```

```
  codeql/java-all: "*"
```

```
EOF
```

```
cd "$Q"
```

```
codeql pack install
```

```
cd -
```

Query 1 direct call to jndilookup.lookup ():

```
cat > "$Q/find-jndilookup-direct-call.ql" <<'EOF'
```

```
/**
```

```
 * @name Direct calls to JndiLookup.lookup
```

```
 * @description Finds direct source-level calls to  
org.apache.logging.log4j.core.lookup.JndiLookup.lookup().
```

```
 * @kind problem
```

```
 * @problem.severity warning
```

```
 * @id java/log4shell/springboot-direct-jndilookup-call
```

```
 */
```

```
import java
```

```
from MethodCall call, Method target
```

```
where
```

```
call.getMethod() = target and
```

```
target.getName() = "lookup" and
```

```
target.getDeclaringType().hasQualifiedName(
```

```
    "org.apache.logging.log4j.core.lookup",
```

```
    "JndiLookup"
```

```
)  
select call, "Direct call to JndiLookup.lookup()."  
EOF
```

```
codeql query run "$Q/find-jndillookup-direct-call.ql" \  
  --database="$DB" \  
  --output="$R/jndillookup-direct-call.bqrs"
```

```
codeql bqrs decode "$R/jndillookup-direct-call.bqrs" \  
  --format=csv \  
  --output="$R/jndillookup-direct-call.csv"
```

```
cat "$R/jndillookup-direct-call.csv"
```

Query 2 direct reference to Jndillookup:

```
cat > "$Q/find-jndillookup-type-reference.ql" <<'EOF'  
  
/**  
  
 * @name Direct references to JndiLookup type  
  
 * @description Finds direct source-level references to  
org.apache.logging.log4j.core.lookup.JndiLookup.  
  
 * @kind problem  
  
 * @problem.severity warning  
  
 * @id java/log4shell/springboot-jndillookup-type-reference  
  
 */
```

```
import java
```

```
from TypeAccess access, RefType t
```

```
where
```

```
access.getType() = t and
```

```
t.hasQualifiedName(
```

```
"org.apache.logging.log4j.core.lookup",
```

```
"JndiLookup"
```

```
)
```

```
select access, "Direct reference to JndiLookup type."
```

```
EOF
```

```
codeql query run "$Q/find-jndilookup-type-reference.ql" \
```

```
--database="$DB" \
```

```
--output="$R/jndilookup-type-reference.bqrs"
```

```
codeql bqrs decode "$R/jndilookup-type-reference.bqrs" \
```

```
--format=csv \
```

```
--output="$R/jndilookup-type-reference.csv"
```

```
cat "$R/jndilookup-type-reference.csv"
```